

Randomness Pattern Hunt Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	randomness-pattern-hunt

Big idea: Human expectations about randomness can mistake clusters for causes and neatness for chance.

One-Minute Teacher Brief

Students compare real and invented random sequences to see why randomness often looks less neat than expected.

Student Prompt

Inspect Randomness Pattern Hunt Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Randomness Pattern Hunt Stimulus A	Student-facing stimulus 1: Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Randomness Pattern Hunt Stimulus B	Student-facing stimulus 2: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Randomness Pattern Hunt Reveal / Twist	Student-facing stimulus 3: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Human expectations about randomness can mistake clusters for causes and neatness for chance.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Randomness Pattern Hunt Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Randomness Pattern Hunt Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	Why do humans see meaning in random patterns?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how human expectations about randomness can mistake clusters for causes and neatness for chance.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Why do humans see meaning in random patterns?

Setup Checklist

- Prepare human-made sequences that avoid long streaks and real random sequences that include clusters.
- Ask students to commit to a judgement before revealing the source.

Example Stimuli or Materials

- Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H.
- Ask: which sequence looks more random, and what makes it feel that way?

Resources To Use

- Resource pack: <resources/randomness-pattern-hunt/index.html>
- Card deck CSV: <resources/randomness-pattern-hunt/cards.csv>
- Visual prompt: <resources/randomness-pattern-hunt/visual-prompt.svg>
- Sequence annotation sheet for streaks, alternations, clusters, and gut judgement.
- Reveal card explaining why random sequences often contain clumps.

Run Sequence

1. Show several short H/T or 0/1 sequences and ask students which look random.
2. Students mark streaks, alternations, and suspicious-looking clusters.
3. Reveal which sequences were random and which were invented.
4. Groups discuss why real randomness often includes streaks.
5. Debrief the difference between pattern perception and probabilistic reasoning.

Debrief Moves

- Knowledge question: Why do humans see meaning in random patterns?
- When should a pattern make us look for a cause?
- How does mathematics discipline intuition?

Teacher Quick Script

- Random does not mean evenly mixed.
- Which pattern made you want to invent a cause?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...